

Chapter 1 Differentials in Complex Field

1.1 Previews

Review:

- Complex Field
- Continuity

1.1.1 Real and Complex Vector Spaces, Complexification

Consider the notation as following:

- $Hom_{\mathbb{R}}(\mathbb{C})$ for $\mathbb{R}$ -linear mappings;
- $Hom_{\mathbb{C}}(\mathbb{C})$ for $\mathbb{C}$ -linear mappings;

Then,

◇ **Lemma 1.1.1.1** Let $T \in Hom_{\mathbb{R}}(\mathbb{C})$ be an $\mathbb{R}$ -linear endomorphisms of $\mathbb{C}$, with following matrix representation for basis $s1, i$

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Then,

$$Tz = \lambda z + \mu \bar{z} \quad \begin{cases} \lambda = \lambda_1 + i\lambda_2 = \frac{1}{2}[(a+d) - (b-c)i] \\ \mu = \mu_1 + i\mu_2 = \frac{1}{2}[(a-d) + (b+c)i] \end{cases}$$

we have: T is $\mathbb{C}$ -linear $\Leftrightarrow a = d, b = -c$.

◦ **Proof** With $z = \begin{pmatrix} x \\ y \end{pmatrix}$ we have

$$Tz = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} = (\lambda_1 + i\lambda_2)(x + iy) + (\mu_1 + i\mu_2)(x - iy) = \begin{pmatrix} \lambda_1 x - \lambda_2 y + \mu_1 x + \mu_2 y \\ \lambda_2 x + \lambda_1 y - \mu_1 y + \mu_2 x \end{pmatrix}$$

Which implies

$$\begin{cases} a = \lambda_1 + \mu_1 \\ b = -\lambda_2 + \mu_2 \\ c = \lambda_2 + \mu_2 \\ d = \lambda_1 - \mu_1 \end{cases} \Rightarrow \begin{cases} \lambda_1 = \frac{a+d}{2} \\ \lambda_2 = -\frac{b-c}{2} \\ \mu_1 = \frac{a-d}{2} \\ \mu_2 = \frac{b+c}{2} \end{cases} \Rightarrow \begin{cases} \lambda = \lambda_1 + i\lambda_2 = \frac{1}{2}[(a+d) - (b-c)i] \\ \mu = \mu_1 + i\mu_2 = \frac{1}{2}[(a-d) + (b+c)i] \end{cases}$$

And $\mathbb{C}$ -Linearity: Notice $id_{\mathbb{C}}$ is linear, but $(\bar{\cdot}) : \mathbb{C} \rightarrow \mathbb{C}$ is not. □

1.2 Complex Differentials, Cauchy-Riemann Equation

1.2.1 Differentiability

Complex Derivative

Definition 1.2.1.1 Define

$$f'(z_0) := \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Then, we have: $f(z) = u(z) + i v(z)$, and notation $u(z) = u(x + iy)$, $v(z) = v(x + iy)$, if $f'(z_0)$ exists,

$$\begin{aligned} f'(z_0) &= \lim_{z \rightarrow z_0} \frac{u(z) - u(z_0)}{z - z_0} + i \lim_{z \rightarrow z_0} \frac{v(z) - v(z_0)}{z - z_0} \\ &= \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{u(z + \Delta x + i \Delta y) - u(z + \Delta x)}{\Delta x + i \Delta y} \\ &\quad + \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \frac{u(z + \Delta x) - u(z)}{\Delta x + i \Delta y} \\ &\quad + i \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{v(z + \Delta x + i \Delta y) - v(z + \Delta x)}{\Delta x + i \Delta y} \\ &\quad + i \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \frac{v(z + \Delta x) - v(z)}{\Delta x + i \Delta y} \\ &= [-i \frac{\partial u}{\partial y} + \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}](z_0) \end{aligned}$$

As matrix coordinate

$$\begin{pmatrix} \frac{\partial u}{\partial x}(z_0) & -\frac{\partial v}{\partial y}(z_0) \\ -\frac{\partial u}{\partial y}(z_0) & \frac{\partial v}{\partial x}(z_0) \end{pmatrix}$$

1.2.2 Cauchy-Riemann Equation

► **Theorem** (Cauchy-Riemann)

$$f'(z) \text{ is } \mathbb{C}\text{-linear} \Leftrightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

◦ **Proof** Omitted. □

1.2.3 Wirtinger Derivation

Define:

Definition 1.2.3.1 With

$$f(z) = f(z_0) + \lambda(z - z_0) + \overline{z - z_0} + o(z - z_0)$$

We have

$$\begin{aligned} \frac{\partial f}{\partial z}(z_0) &= \lambda = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \\ \frac{\partial f}{\partial \bar{z}}(z_0) &= \mu = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \end{aligned}$$

♠ **Proposition 1.2.3.2** We have:

- If f is at z_0 (real)-differentiable, then

$$f(z) = f(z_0) + \frac{\partial f}{\partial z}(z_0)(z - z_0) + \frac{\partial f}{\partial \bar{z}}(z_0)\overline{(z - z_0)} + o(z - z_0) \quad z \rightarrow z_0$$

- $\frac{\partial}{\partial z}(f \cdot g) = \frac{\partial f}{\partial z} \cdot g + f \cdot \frac{\partial f}{\partial z};$

- We have:

$$\frac{\partial f}{\partial z} = \overline{\frac{\partial \bar{f}}{\partial \bar{z}}} \quad \frac{\partial \bar{f}}{\partial \bar{z}} = \overline{\frac{\partial f}{\partial z}}$$

- For $f : z \mapsto z, g : z \mapsto \bar{z}$ we have

$$\frac{\partial}{\partial z} f = 1; \frac{\partial}{\partial \bar{z}} f = 0; \frac{\partial}{\partial z} g = 1; \frac{\partial}{\partial \bar{z}} g = 0$$

- (Chain-Rule I) Suppose $g(w), f(z)$ continuous differentiable, we have

$$\frac{g \circ f}{\partial z} = \frac{\partial g}{\partial w} \circ$$

- (Chain-Rule II) Suppose $\varphi : [a, b] \rightarrow \mathbb{C}$ continuous differentiable, then

$$\frac{d}{dt}(f \circ \varphi) = \frac{df}{dz} \circ \varphi \cdot \frac{d\varphi}{dt} + \frac{d}{d\bar{z}} \circ \varphi \cdot \overline{\frac{d\varphi}{dt}}$$

◦ **Proof**

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□

1.2.4 Complex Differentiability

♠ **Proposition 1.2.4.1** Suppose $f : U \rightarrow \mathbb{C}$ be a function with $U \subseteq \mathbb{C}$ open. The followings are equivalent:

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◦ **Proof**

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□

Definition 1.2.4.2 (Holomorphic Functions)

- f is **holomorphic** on U , if f is complex-differentiable at each $z \in U$;
- Take the notion of $\mathcal{O}(U)$ for the collection of holomorphic functions.

♠ **Proposition 1.2.4.3** (Rules of Complex Derivative)

- Suppose f, g complex differentiable at z_0 , then at the point z_0 , we have

...

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◦ **Proof** ...

□

1.2.5 Polynomials, Power Series

Polynomials

♠ **Proposition 1.2.5.1** Polynomial $p \in \mathbb{C}[z]$ is complex differentiable, and

$$p'(z) = \sum_{j=1}^n j a_j z^{j-1}$$

◦ **Proof** Omitted.

□

Power Series

Definition 1.2.5.2 Define the power series:

$$f(z) = \sum_{i=0}^{\infty} a_i (z - z_0); z_0 \in \mathbb{C}$$

We have

♠ **Proposition 1.2.5.3** (Cauchy-Hadamard) The convergence radius

$$R = \frac{1}{\limsup |c_n|^{\frac{1}{n}}}$$

where:

- $|z| > R$ Divergent;
- $|z| = R$ Convergent or Divergent;
- $|z| < R$ Convergent.

◦ **Proof**

- $|z| > R$,

$$\sqrt[n]{|c_n|}$$

- $|z| < R$,

$$\sqrt{n}|c_n| \geq t + \epsilon$$

for finite of n . And for others,

$$\sqrt{nc_n} < t + \epsilon \Rightarrow$$

□

★ **Corollary 1.2.5.4** If the radius R and $\lim \left| \frac{a_{n+1}}{a_n} \right|$ exists, then,

$$R = \lim \left| \frac{a_{n+1}}{a_n} \right|$$

Some of review in analysis

♠ **Proposition 1.2.5.5** If $\{a_n\}, \{b_n\}$ is absolute convergent, then definen $c_n := \sum_{k=0}^n a_k b_{n-k}$,

$$\sum_{n=0}^{\infty} c_n = \left(\sum a_n \right) \left(\sum b_n \right)$$

◦ **Proof ...**

□

1.3 Fundamental Theorems in Complex Analysis

1.3.1 Invertible Function

♠ **Proposition 1.3.1.1** Suppose $f : U \rightarrow V$ continuous continuous complex differentiable, bijective. If for every $z \in U$,

$$f'(z) \neq 0$$

Then, there exists an inverse $f^{-1} : V \rightarrow U$ with

$$(Df^{-1}(f(z))) = (Df(z))^{-1}, z \in U$$

◦ **Proof ...**

□

♠ **Proposition 1.3.1.2** Suppose $f : U \rightarrow V$ be continuous complex differentiable, with $U, V \subseteq \mathbb{C}$ open. And f is bijective, $f'(z) \neq 0, z \in U$.

Then, the inverse $f^{-1} : V \rightarrow U$ is also continuous complex differentiable, with

$$(f^{-1})'(w)f'(f(w)) = 1$$

◦ **Proof Omitted.**

□

1.3.2 Analytic Branch

Logarithms

Definition 1.3.2.1 Define:

- $C_\alpha = \mathbb{C} - \{re^{i\alpha} | r \geq 0\}, 0$
- $\Omega_\alpha = \{z \in \mathbb{C} | \alpha - 2\pi < \text{Im}z < \alpha\}$

We have:

- Exponential

$$\exp : \Omega_\alpha \rightarrow C_\alpha$$

- Logarithms

$$\log_\alpha : C_\alpha \rightarrow \Omega_\alpha$$

Especially, logarithm $\log_\pi$ is called **main branch**.

Roots

Definition 1.3.2.2 Define:

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$$z \mapsto z^n \quad \{z \in \mathbb{C}^* | 0 \leq |\arg(z)| \leq \pi/n\} \rightarrow \mathbb{C}_\pi$$

which is bijective.

- Then, the inverse

$$\sqrt[n]{-} : z \mapsto \sqrt[n]{z}$$

1.3.3 Complex Integration

Area Integraion

As what we did in $\mathbb{R}^2$,

Path Integration

Define the path integration as following:

$$\int_\gamma f(z)dz = \lim_{\Delta \rightarrow 0} \sum f(z_i)\Delta z_i$$

And we have the coordinate decomposition

$$f(z) = u(z) + iv(z) \quad dz = dx + idy$$

Finally:

$$= \int_\gamma (u + iv)(dx + idy) = \int_\gamma udx - vdy + i \int_\gamma vdx + udy$$

Especially: If the path γ is continuous, and $\gamma|[t_i, t_{i+1}] \in C^1([t_i, t_{i+1}])$, we have

$$\int_\gamma f(z)dz = \int_a^b (f \circ \gamma)(t) \cdot \gamma'(t)dt = \sum \int_{t_{i-1}}^{t_i} f(\gamma(t)) \cdot \gamma'(t)dt$$

♣ Example 1.3.3.1

♠ Proposition 1.3.3.2 ...

◦ Proof ...

□

'Holzhammer Ungleichung'

♠ Proposition 1.3.3.3 ...

◦ Proof ...

□

♠ Proposition 1.3.3.4 (Main Theorem of Calculus for Path Integration in Complex Analysis)

$$\int_{\gamma} F'(z) dz = F(\gamma(b)) - F(\gamma(a))$$

◦ Proof ...

□

1.3.4 The Winding Numbers

Definition 1.3.4.1 For a path γ with $a \notin sp\gamma$, define

$$wind(\gamma, a) = \frac{1}{2\pi i} \frac{d\gamma}{z - a}$$

◇ **Lemma 1.3.4.2** $f : U \rightarrow \mathbb{C}$ continuous complex differentiable, with γ a path in U , then,

$$\exp\left(\int_{\gamma} \frac{f'}{f}(z) dz\right) = \frac{f(\gamma(b))}{f(\gamma(a))}$$

◦ **Proof** Consider

$$\Phi(t) = \exp\left(-\int_a^t \frac{f'}{f}(\gamma(s)) \gamma'(s) ds\right) \cdot f(\gamma(t))$$

We have:

$$\Rightarrow \Phi(a) = \Phi(b), \exp\left(\int_{\gamma} \frac{f'}{f}(z) dz\right) = \frac{f(\gamma(b))}{f(\gamma(a))}$$

□

♠ Proposition 1.3.4.3

$$\oint_{|\gamma-a|=r} \frac{d\xi}{(\xi - z)^n} = \begin{cases} 0 & n \neq -1 \\ 2\pi i & n = -1 \end{cases}$$

◦ **Proof** Now consider $\exp()$

□

1.3.5 Cauchy's Formula

Triangle

► **Theorem** (Cauchy) If f is analytic on $\Omega \subseteq \mathbb{C}$, with triangle $\Delta ABC \subseteq \Omega$,

◦ **Proof** ...

□

Star Region

Definition 1.3.5.1 (Star Region)

Stroke $\Rightarrow$ Cauchy's Integration Formula

1.3.6 Mean Value Property

♠ **Proposition 1.3.6.1** Suppose f is analytic in $B(z_0, R)$; $R > r$ then,

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta$$

◦ **Proof** ...

□

★ **Corollary 1.3.6.2** We have

$$|f(z_0)| = \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{i\theta})| d\theta$$

◦ **Proof** ...

□

1.3.7 Argument Principle

Suppoe that $f(z) = (z - a)^m g(z)$, with m th zero pole, then

$$f'(z) = (z - a)^m g'(z) + m(z - a)^{m-1} g(z); m \geq 1 \Rightarrow \frac{f'(z)}{f(z)} = \frac{g'(z)}{g(z)} + \frac{m}{z - a}$$

Conversely, suppose $f(z) = (z - a)^{-m} g(z)$ with pole of order m , then

$$f'(z) \Rightarrow$$

► **Theorem** (Argument Principle) Let

$$\frac{1}{2\pi} \int \frac{f'(z)}{f(z)} dz = \sum n(\gamma; z_i) - \sum n(\gamma, p_i)$$

◦ **Proof ...**

□

1.3.8 Rouché's Theorem

► **Theorem** (Rouche) Suppose D be a region on $\mathbb{C}$, with a

1.3.9 Open Mapping's Theorem

► **Theorem** Suppose f is a non-constant analytic function on the region Ω , then its image

$$f(\Omega)$$

is open.

◦ **Proof ...**

□

1.4 Singularities

1.4.1 Classification

Isolated Singularities

Definition 1.4.1.1 A function f has an **isolated singularity** at $a \in \mathbb{C}$, if there exists $R > 0$ with

- f is analytic on $B(a, R) - a$;
- f is not analytic on $B(a, r)$;

Classification of Isolated Singularities:

- f is called **removable** if
- f is called **text** if

1.4.2 Residue's Theorem

► **Theorem** (Residue's Theorem)

◦ **Proof ...**

□

1.5 The Maximum and Minimum Modulus Theorem

1.5.1 Maximum Modulus Theorem

► **Theorem** (Maximum Modulus Theorem, the First Version) Suppose G be a region, with f analytic on G . $a \in G$ with

$$|f(z)| \leq |f(a)| \quad z \in G$$

Then, f is constant.

◦ Proof ...

□

► **Theorem** (Maximum Modulus Theorem, the Second Version)

◦ Proof ...

□

► **Theorem** (Maximum Modulus Theorem, the Third Version)

◦ Proof ...

□

Another proof can also be given as following:

Statement The harmonic function has the maximum/minimum on the boundary.

1.5.2 Minimum Modulus Theorem

► **Theorem** ...

1.5.3 Schwarz's Theorem

1.5.4 Hadamard's Three Circle Theorem

1.6 Compactness and Convergence in the Space of Analytic Functions

1.6.1 The Riemann's Mapping Theorem

► **Theorem** ...

◦ Proof ...

□

More topics will be given in: Theory of Univalent Functions.

Topic: Biberbach's Polynomial

◇ Lemma 1.6.1.1 ...

◦ Proof ...

□

1.7 Weierstrass's Factorization Theorem

► Theorem ...

◦ Proof ...

□

1.8 Runge's Theorem

1.9 Analytic Continuation

Original Riemann Surface: Solution of algebraic function

$$f(w, z) = 0$$

Especially: If $f(w_0, z_0) = 0$; $f_z(w_0, z_0) \neq 0 \Rightarrow$ Locally $z = z(w)$.**Definition 1.9.0.1**

- Suppose $\gamma : [0, 1] \rightarrow \Omega$, then $D_j(\gamma(t_i), r_i) \subseteq \Omega$ is a **chain of disks** iff
 - There exists cut $0 = t_1 < \dots < t_N = 1$,

$$\gamma([t_i, t_{i+1}]) \subseteq D_i \cap D_{i+1}$$

- $g \in \mathcal{H}(V)$ is an **analytic continuation** of f if there exists a chain of disks with $f_j = f_{j+1}$ on $D_j \cap D_{j+1}$ and

$$f_0 = f \quad f_J = g$$

◇ **Lemma 1.9.0.2** The analytic continuation g of f along γ only depends of f and γ , but not on the choice of circles. In particular, it is unique.

◦ Proof ...

□

► **Theorem** Suppose γ_0, γ_1 two homotopic curves relative to some region $\Omega \subseteq \mathbb{C}$, with initial and end points p, q . Let U be a neighborhood of p , assume $f \in \mathcal{H}(U)$, then the analytic continuation agrees locally around p, q .

◦ Proof ...

□

Remark

Example: $\Gamma(z)$

Consider

$$\Gamma(z) := \int_0^\infty e^{-t} t^{z-1} dt$$

It is analytic on the half plane:

Chapter 2 Harmonic Functions

Chapter 3 Entire Functions

3.1 Convergence, and Normal Families

◇ **Lemma 3.1.0.1** Suppose

$$\{f_n\}_{n=1}^{\infty} \subseteq \mathcal{H}(\Omega)$$

converges uniformly on compact subsets of Ω to a function f , then

- $f \in \mathcal{H}(\Omega)$;
- $f_n^{(k)} \rightarrow f^{(k)}$ uniformly on compact subset of Ω for each $k \geq 1$.

Furthermore, suppose

$$\sup_{n \geq 1} \#\{z \in \Omega | f_n(z) = w\} \leq N \leq \infty$$

Then, either $f \equiv w$ in Ω or $\#\{z \in \Omega | f(z) = w\} \leq N$.

Remark The cardinalities include multiplicity.

◦ **Proof ...**

□

Definition 3.1.0.2 (Normal Family) One says that $\mathcal{F} = \{f_\alpha\}_{\alpha \in A} \subseteq \mathcal{H}(\Omega)$ is a **normal family** provided for each compact $K \subseteq \Omega$ one has

$$\sup_{z \in K} \sup_{\alpha \in A} |f_\alpha(z)| < \infty$$

♠ **Proposition 3.1.0.3** One says: $\mathcal{F} = \{f_\alpha\}_{\alpha \in A} \subseteq \mathcal{H}(\Omega)$ is a **normal family**. Then, there exists a sequence f_n in $\mathcal{F}$ that converges uniformly on compact subset of Ω . The limit again being a holomorphic function.

◦ **Proof ...**

□

► **Theorem** Let $\mathcal{F}$ be a family of meromorphic functions in Ω , such that for each $z_0 \in \Omega$ there is a neighborhood $U(z_0) \subseteq \Omega$ and constant $M(z_0)$ such that:

- Either $|f(z)| < M(z_0)$;
- Or $\frac{1}{|f(z)|} < M(z_0), \forall z \in U(z_0), \forall f \in \mathcal{F}$

Then, $\mathcal{F}$ is normal.

◦ **Proof ...**

□

3.2 Mittag-Lefflers Theorem

Review: Mittag-Leffers system, HA

► **Theorem** (Mittag-Lefflers Theorem) Given $\{z_n\}_{n=1}^N \subseteq \mathbb{C}$, with $|z_n| \rightarrow \infty$ if $N \rightarrow \infty$, and polynomial P_n with positive degrees and no constant terms, there exists $f \in \mathcal{M}(\mathbb{C})$ so that f has poles exactly at z_n , and

$$f(z) - P_n\left(\frac{1}{z - z_n}\right)$$

is analytic around z_n for each $1 \leq n \leq N$.

◦ **Proof ...** □

★ **Corollary 3.2.0.1** Consider the following exact sequence

$$0 \rightarrow \mathcal{O} \rightarrow \mathcal{M} \rightarrow \mathcal{M}/\mathcal{O} \rightarrow 0$$

which induces a long sequence:

$$0 \rightarrow H^0(X, \mathcal{O}) \xrightarrow{i} H^0(X, \mathcal{M}) \xrightarrow{\pi} H^0(X, \mathcal{M}/\mathcal{O}) \xrightarrow{\delta} H^1(X, \mathcal{O}) \rightarrow \dots$$

where $X \subseteq \mathbb{C}$ is a domain. Now, we have:

$$H^1(X, \mathcal{O}) = 0$$

◦ **Proof ...** □

3.3 Runge's Theorem

3.3.1 Runge's Theorem

Firstly: If $f \in \mathcal{O}(U)$, then it is analytic, and has an approximation

$$f = \lim_n \sum a_i(z - z_0)^i; z \in D(z_0, r)$$

But a question is:

If we can replace disk $D(z_0, r)$ with **arbitrary region**?

The answer is **no**:

♣ **Example 3.3.1.1** Consider $G = \{z \in \mathbb{C} | 0 < |z| < 1\}$. Then, if $p_n \rightarrow f$, then consider the path

$$\gamma = e^{it}$$

We have

$$\lim \oint_{\gamma} p_n = \oint_{\gamma} \lim p_n = \oint_{\gamma} f = 0$$

But, $\oint z^{-1} = 0$, and z^{-1} is analytic on G . The main reason is simply-connectedness.

◇ **Lemma 3.3.1.2** Suppose K a compact subset of the region G , then there are upper line segments

$$\gamma_1, \dots, \gamma_n \subseteq G - K$$

such that for every function f in G ,

$$f(z) = \sum \frac{1}{2\pi i} \int_{\gamma_k} \frac{f(w)}{w - z} dw$$

◦ **Proof ...**

□

◇ **Lemma 3.3.1.3** $B(E)$ is a closed subalgebra of $C(K, \mathbb{C})$ that contains every rational function with a pole in E .

◦ **Proof ...**

□

◇ **Lemma 3.3.1.4** ...

◦ **Proof ...**

□

◇ **Lemma 3.3.1.5** ...

◦ **Proof ...**

□

► **Theorem** Let K compact subset of $\mathbb{C}$, and E a subset of $\mathbb{C}_\infty - K$.
If:

- E meets each (connected) components of $\mathbb{C}_\infty - K$;
- f is analytic in an open set containing K ;
- $\epsilon > 0$

Then, there exists a rational number $R(z)$, whose only pole lie in E , and such that

$$|f(z) - R(z)| < \epsilon$$

◦ **Proof ...**

□

Remark Here, we could give another proofs:

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3.3.2 The Density of Functions

3.3.3 Polynomially Convex

Definition 3.3.3.1 ...

Chapter 4 The Range of Analytic Functions

Continuation of Analytic Functions, given in Riemann Surface.